

Problem A. All Closed

Time Limit: 1 seconds | Memory Limit: 512 MB

Standard Input / Standard Output

Sulfox the fennec fox gives you n sets $S_1, S_2, \dots, S_n$, each containing distinct integers between 0 and $2^m - 1$.

You may perform the following operation any number (possibly zero) of times: choose an integer x with $0 \leq x < 2^m$, and for every $i = 1, 2, \dots, n$, insert x into S_i if it is not already present.

A set S is called **XOR-closed** if, for every $x, y \in S$, we have $x \oplus y \in S$, where $\oplus$ denotes the bitwise exclusive-OR operation.

Find the minimum number of operations required to make all S_i XOR-closed, and construct any sequence of operations achieving this minimum.

Input

The first line contains two integers n ($1 \leq n \leq 2 \times 10^5$) and m ($1 \leq m \leq 20$).

The i -th of the next n lines contains an integer c_i ($c_i \geq 1$), followed by c_i distinct integers $a_{i,1}, a_{i,2}, \dots, a_{i,c_i}$ ($0 \leq a_{i,j} < 2^m$), representing the elements of S_i .

It is guaranteed that the sum of c_i does not exceed 10^6 .

Output

On the first line, print the minimum number of operations k .

On the second line, print k integers $x_1, x_2, \dots, x_k$, where x_i denotes the integer chosen in the i -th operation.

If there are multiple optimal solutions, output any of them.

Examples

Sample Input 1	Sample Output 1
3 3 1 1 3 0 1 7 6 1 2 3 4 5 7	3 0 7 6
Sample Input 2	Sample Output 2
1 4 4 11 5 14 0	0

Note

For the first example, after performing the shown operations, S_1 and S_2 become $\{0, 1, 6, 7\}$, while S_3 becomes $\{0, 1, 2, 3, 4, 5, 6, 7\}$. It can be proved that it is impossible to make all sets XOR-closed using less than 3 operations.

For the second example, the only set is already XOR-closed, so no operation is needed.